Graphic Era HILL UNIVERSITY
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University under section 2(f) of UGC Act, 1956

Mid Term (Odd) Semester Examination October 2024

Roll no.....

Name of the Course and semester:
Name of the Paper: Refrigeration & Air Conditioning
Paper Code: TME 506
Time: 1.5 hour

Maximum Marks: 50

Note:

(i) Answer all the questions by choosing any one of the sub questions
(ii) Each question carries 10 marks.

Q1. (10 Marks)

a. Derive the efficiency expression for the reverse Carnot cycle and discuss its significance in refrigeration applications. **CO 1**

OR

b. Explain the working principle of the Bell Coleman cycle used in aircraft refrigeration. Include a schematic diagram and discuss each component's role in the cycle. **CO 1**

Q2. (10 Marks)

a. A gas refrigeration system operates on the Bell Coleman cycle. The refrigeration capacity is 10 kW, and the temperature of the cold reservoir is -10°C. The pressure ratios across the compressor are 4:1. Calculate the work input to the compressor and the coefficient of performance (COP) of the system. **CO 1**

OR

b. A refrigeration system operates on a reverse Carnot cycle. The refrigerant is air, and the system operates between two reservoirs, with the evaporator maintained at −20°C and the condenser at 30°C. The system provides a refrigeration effect of 50 kW.

i. Calculate the coefficient of performance (COP) of the system.
ii. Determine the power input required by the system.
iii. Calculate the total heat rejected to the surroundings. **CO 1**

Q3. (10 Marks)

a. List and explain the advantages of using gas refrigeration systems over conventional refrigeration methods. Provide specific examples to illustrate your points. **CO 1**

OR

b. Explain the working of a multi-stage compression system in refrigeration. **CO 2**

Q4. (10 Marks)

a. A vapour compression cycle uses R-12 as a refrigerant with a refrigeration capacity of 50 kW. The evaporator operates at -5°C and the condenser at 40°C. Determine:
(a) The mass flow rate of the refrigerant (kg/s),
(b) The compressor power (kW),
(c) The COP of the cycle. Use appropriate refrigerant tables. **CO 2**

OR

b. In a simple vapour compression refrigeration system, the refrigerant R-22 enters the compressor at saturated vapour state at -10°C and exits the condenser at 40°C. The mass flow rate of refrigerant is 0.08 kg/s. Calculate:
(a) The compressor work (kW),
(b) The refrigeration effect (kW),
(c) The COP of the system. **CO 2**

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Q5.

(10 Marks)

a. Explain the different methods of improving the COP of a vapour compression refrigeration system. **CO 2**

OR

b. Explain the effect of evaporator and condenser pressures on the performance of the refrigeration cycle. How do changes in these pressures influence the COP and refrigeration effect? **CO 2**

Refrigerant Properties for R-12

State	Temperature (°C)	Pressure (kPa)	Enthalpy (h, kJ/kg)	Entropy (s, kJ/kg·K)	Specific Volume (v, m³/kg)
Saturated Liquid (h_f)	-10	193.8	183.54	0.926	0.000875
Saturated Vapour (h_g)	-10	193.8	373.68	1.723	0.1066
Saturated Liquid (h_f)	40	808.3	209.26	0.994	0.001182
Saturated Vapour (h_g)	40	808.3	404.78	1.739	0.0525
C_p (liquid) = 1.195 kJ/kg·K					
C_p (vapour) = 0.629 kJ/kg·K					

Refrigerant Properties for R-22

State	Temperature (°C)	Pressure (kPa)	Enthalpy (h, kJ/kg)	Entropy (s, kJ/kg·K)	Specific Volume (v, m³/kg)
Saturated Liquid (h_f)	-10	402.5	233.68	1.065	0.00095
Saturated Vapour (h_g)	-10	402.5	405.12	1.718	0.0546
Saturated Liquid (h_f)	40	1610	267.05	1.255	0.0014
Saturated Vapour (h_g)	40	1610	432.32	1.761	0.0347
C_p (liquid) = 1.435 kJ/kg·K					
C_p (vapour) = 0.662 kJ/kg·K					

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